

Keeper batch tier-gate — Lyra three new items: bulk-color v0.6 (INTERNAL CORRECTION of v0.5; closes Keeper C3 SO(3)-decomposition concern via Cartan-Weyl basis $3+3+2 = 3 T_a + 3 T_a^\dagger + 2 K$ -Cartan), Spectral Score v0.2 (Saturday absorptions synthesis), T1 Falsifier v0.2 (CD-layering sharpening + sensitivity matrix + composite scenarios). All three PASS. v0.6 is the substantive close on my v0.5 C3 concern; v0.2 partially addresses my v0.1 N1 notes (4th-gen + extra-K-type triggers via sensitivity matrix).

Keeper

2026-05-30 Saturday late-morning

Batch tier-gate — Lyra v0.6 + Spectral Score v0.2 + T1 v0.2

1. Lyra Bulk-color v0.6 — Internal Correction of v0.5

Verdict: PASS at CANDIDATE FRAMEWORK with C3 SUBSTANTIVELY CLOSED. This is the right kind of internal correction — Lyra independently caught the same SO(3)-decomposition mismatch I flagged in my v0.5 tier-gate (C3 condition) and corrected via the Cartan-Weyl basis.

The correction: - v0.5 (now retired): $8 = 3 T_a + 3 [T_a, T_b] + 2 K$ -Cartan. STRUCTURALLY MISMATCHED for su(3) (the $[T_a, T_b]$ commutators don't play the role of negative-root vectors ($e_{-\alpha}, f_{-\alpha}$) — they would be additional positive root vectors of a higher-dim algebra). - v0.6 (revised): $8 = 3 T_a + 3 T_a^\dagger + 2 K$ -Cartan — Cartan-Weyl basis: 3 raising operators + 3 lowering operators + 2 Cartan. Structurally aligned with su(3).

This SUBSTANTIVELY CLOSES my C3 condition from the v0.5 tier-gate (which read: “v0.6 must (a) identify non-canonical SO(3) embedding, (b) verify non-canonical su(3) closure, OR (c) honestly mark as count-aligned-OPEN”). Lyra's v0.6 takes the cleaner route: revise the structural assignment to match su(3) standard Cartan-Weyl, with $T_a^\dagger$ as the negative-root partners. This is option (a) at a different layer — the Toeplitz operators T_a and their Hermitian conjugates $T_a^\dagger$ are the natural raising/lowering pair under the substrate-Toeplitz algebra structure.

Strengths: - Cal #27 self-applied internally (“peak-convergence overclaim caught”); v0.5 → v0.6 within ~1 hour is exemplary self-correction discipline. - v0.6 correctly identifies what v0.5 was right about (Toeplitz algebra non-commutativity from Hardy projection; SO(3) ⊂ SO(5) sub-vector 3-fold; numerical 8-count) and what v0.5 was wrong about (the structural decomposition assignment). - The $[T_a, T_b]$ commutators are now correctly understood as DERIVED nonzero compact operators in the Toeplitz algebra, not as independent generators — they're consequences of the algebra closure, not new fundamentals. - The revised candidate is structurally honest at the count + organization level.

Remaining open burdens (v0.7+ pending Elie's lane): - C4 carried forward (from v0.5): explicit Toeplitz commu-

tator computation: - $[T_a, T_a^\dagger] = \alpha(H_?)$ — does this give substrate-natural Cartan element matching $\mathfrak{su}(3)$ roots? - $[T_a, T_b] = N_{\{\alpha\beta\}} T_{\{\alpha+\beta\}}$ — does the root-sum structure close into another T-operator? - C5 (NEW): hermiticity, semisimplicity, traceless conditions — does the resulting algebra satisfy $\mathfrak{su}(3)$ (vs $\mathfrak{u}(3)$ or some other 8-dim algebra)? - C1 (Toeplitz-source citation) + C2 (AdS/CFT demotion) carry from v0.4.

Tier disposition: - Internal correction discipline (Section 0-1, 3): EXEMPLARY — Cal #27 working on Lyra's own work in real time - Cartan-Weyl basis $3 + 3 + 2$ structural assignment: CANDIDATE STRUCTURALLY ALIGNED with $\mathfrak{su}(3)$; count match remains 8 - Closure verification: OPEN multi-week (C4 + C5) - C1+C2: carry from v0.4 (Toeplitz source verification + AdS/CFT demotion still needed in v0.7+)

Status: PASS at CANDIDATE FRAMEWORK; C3 closed via v0.6 correction; C4+C5 remain open multi-week burdens; v0.6 is the substantively-honest closure-pending candidate.

2. Lyra Spectral Score v0.2 — Saturday absorptions

Verdict: PASS at SYNTHESIS with one note: still references v0.5's $3+3+2$ decomposition with $[T_a, T_b]$, not v0.6's corrected $3+3+2$ with $T_a^\dagger$.

Strengths: - Absorbs 5 Saturday substantive pieces (Elie Toy 3627 radial towers; Grace Pair α channels; bulk-color v0.5 8-count; Quasi-Eigentone v0.2 Resolution B; Mersenne-Hall connection) into a richer V0 narrative. - Sector radial towers explicit per Elie Toy 3627 closed forms. - Generation channel labels per Grace Pair α + Resolution B mapping. - Gluon octet sub-structure visualization with substrate-primary anchors. - Mersenne-Hall integer connection adds substrate-numerics narrative. - 9-frame movie narrative refined.

Note for v0.3: - N1 — Update gluon octet visualization to v0.6 Cartan-Weyl $3+3+2$ decomposition ($3 T_a + 3 T_a^\dagger + 2 K$ -Cartan), NOT v0.5's $[T_a, T_b]$ version. The v0.2 doc was filed at 10:45 EDT, before v0.6 correction at 11:30 EDT — staleness of a few hours. Mechanical update in v0.3.

Tier disposition: - Sector radial tower visualization (Section 1): FRAMEWORK (Elie Toy 3627 RIGOROUS) - Generation channel labels (Section 2): FRAMEWORK with Resolution B preferred (Quasi-Eigentone v0.2) - Gluon octet sub-structure (Section 3): FRAMEWORK but stale to v0.5 — N1 update needed - Mersenne-Hall substrate-primary anchors (Section 4): FRAMEWORK - T190 closed form visualization (Section 5): FRAMEWORK consistent with Resolution B - 9-frame movie narrative (Section 6): ORGANIZATIONAL

Status: PASS as Movie V0 organizational spine v0.2; N1 v0.3 mechanical update absorbing v0.6 correction.

3. Lyra T1 Falsifier v0.2 — CD-Layering Sharpening

Verdict: PASS at SHARPENED PREDICTION. Adds quantitative thresholds, sensitivity matrix, composite scenarios, honest noise floors — operationalizes v0.1's falsifier design substantively.

Strengths: - Quantitative thresholds per channel (Section 1): explicit 2σ trigger thresholds (PMNS deviation > 0.003 on θ_{12} ; > 0.010 on θ_{23} ; > 0.0006 on θ_{13}). - Sensitivity matrix (Section 2): maps each potential trigger to what specifically would fail (per-particle PMNS, lepton-sector mapping, keystone bet itself, σ_{BF} parity, K-type catalog, T190 mechanism, GUT, etc.). Partially addresses my v0.1 N1 (4th-gen + extra-K-type triggers) by giving structural disambiguation per trigger. - Composite scenarios (Section 3): multiple weak hits across channels → combined statistic protocol. Addresses the “what if multiple weak signals” case rather than only single-decisive triggers. - Honest noise floors (Section 4): per-channel measurement precision + future tightening (3-10x over 5-10 yr). - Cal #27 self-applied: “v0.2 makes the falsifier design quantitatively concrete ... without overclaiming any current measurement.”

Notes carrying forward (from v0.1, partially addressed): - N1 (4th-gen trigger specification): partially addressed via Section 2 sensitivity matrix entry “ $h^v \neq 3$ generation-count mechanism wrong” — but doesn't explicitly require “doublet-organized Dirac” 4th gen as I'd noted. v0.3 could sharpen further if Cal cold-read flags. - N2 (extra-K-type signature): partially addressed via Section 2 “K-type catalog incompleteness — could be Five-Absence-prediction failure.” Vague but acceptable for the level of design specificity.

Tier disposition: - Quantitative thresholds (Section 1): RIGOROUS (PDG + projected experiment precisions)
 - Sensitivity matrix (Section 2): FRAMEWORK (mapping) - Composite scenarios (Section 3): FRAMEWORK (protocol) - Noise floors (Section 4): RIGOROUS (measurement physics) - Overall falsifier disposition (Section 5): SHARPENED PREDICTION

Status: PASS at SHARPENED PREDICTION; partially addresses my v0.1 N1+N2 notes via sensitivity matrix; Cal cold-read remains the next step for further refinement.

4. Cross-cutting observations

The bulk-color v0.5 $\rightarrow$ v0.6 correction parallels exactly the discipline pattern I'd been calibrating against this week: - Cal #27 fires hardest at peak-convergence overstatement (the 8-count match was peak-convergence) [?] - Calibration #34 STANDING (narrow framing, headline-cap-conditionality) — v0.5 headline carried “first SU(3) count match” which would have stuck even with body-level honest caveats; v0.6 explicitly retires that headline framing [?] - Sustained internal correction discipline — v0.5 $\rightarrow$ v0.6 in ~1 hour without external Cal cold-read is the team's R3 sustainable cadence + Cal #27 STANDING discipline working in real time [?]

This is exactly the cadence + discipline I've been documenting. The Saturday Lyra v0.6 internal correction is a substantive instance of the methodology infrastructure working as designed.

For the Spectral Score v0.2 staleness on v0.5 vs v0.6: this is a small instance of Calibration #34 propagation — when a load-bearing decomposition changes (v0.5 $\rightarrow$ v0.6), downstream visualization docs need synchronized update. v0.3 mechanical fix.

5. Routed handback

- Lyra: Spectral Score v0.3 absorbs N1 (v0.6 corrected 3+3+2 with $T_a^{\dagger}$); T1 v0.3 absorbs Cal cold-read if/when it comes; bulk-color v0.7 pending Elie Toeplitz commutator verification + Cal cold-read on v0.6 closure question.
- Elie: explicit Toeplitz commutator computation on $H^2(S)$:
 - $[T_a, T_a^{\dagger}] = \alpha(H_?) \leftrightarrow$ substrate-natural Cartan element?
 - $[T_a, T_b] = N_{\{\alpha\beta\}} T_{\{\alpha+\beta\}} \leftrightarrow$ root-sum structure?
 - Hermiticity + semisimplicity + tracelessness check for su(3) vs u(3) distinction
 - Multi-week computational task; this is THE bulk-color closure work.
- Cal: cold-read v0.6 (verify Cartan-Weyl basis structural alignment; flag any remaining issues) + cold-read T1 v0.2 (refine sensitivity matrix per referee-question framing).
- Grace: G13 8-gluon scaffold v0.2 should update to v0.6 3+3+2 with $T_a^{\dagger}$ decomposition (parallel to Spectral Score v0.3 N1).

6. Saturday Keeper cumulative

With these tier-gates, Saturday Keeper batch: - ~25 substantive tier-gate / correction / consolidation docs filed - 3 falsifiers LIVE (F1 PMNS, T1 lepton-count, Bell sub-Tsirelson 1/8) - Bulk-color advance: v0.4 COMMITMENT $\rightarrow$ v0.5 COUNT MATCH $\rightarrow$ v0.6 CARTAN-WEYL STRUCTURAL ALIGNMENT — substantive progress through the day; multi-week closure verification remains - Quasi-Eigentone v0.2 Resolution B SYNTHESIS with two-structures-burden flag - Strong-Uniqueness v1.1 STRENGTHENED + null-model recompute (single-residual framing) - Two-Route Scan Cal #171 relation-reduction brake absorbed - Periodic Table v0.7 (6-tuple per-cell with stability class) - Calibration #34 STANDING (narrow framing, headline-cap-conditionality) - Methodology stack 19 layers; discipline stack 5 elements - Vol 16 §3 + §4 + OUTLINE staleness routing

Ready for depth shift per Casey directive. Next concrete options for me: - SP-30 outreach pre-drafts (eigentone, Cs-137, W-32) per Task #330 — substantive proactive but needs Casey sign-off before sending - CLAUDE.md headline status refresh to current Saturday afternoon state — persistent context value for future sessions

— Keeper. v0.6 correction substantively closes C3; Spectral Score v0.2 + T1 v0.2 PASS with minor notes; Saturday batch complete on team-filed items; ready for depth shift.